Assignment - 3

Assigned: April 17, 2026 Due: April 27, 2026 Max. marks: 100

General Instructions

- Submit a hard-copy of the solutions either in-class or in TAs office NAC-2, Room no. 311.
- Upload a scanned copy of the solutions and the codes you have developed as a single zipped folder on Moodle.
- Do not email your solutions/codes. Emailed submissions will not be graded.

Hand Calculations

1. (20 points) In a particular MPI program it was noted that the inherently sequential part of the computation varies as, $\sigma(n) = an + b$, where $a = 10$ and $b = 1000$, and the parallelizable part of the computation varies as, $\phi(n) = cn^3 + dn^2 + en$, where $c = 10^{-3}, d = 0.2$ and $e = 100$ and n is the problem size which can be taken as $n = 1000$. The communication overheads are incurred at $[\log_{10} n]$ locations in the program and each communication consumes $\left[1000(\log_{10} p) + (n/10)\right]$, where p is the number of processes and can be taken as $p = 100$. Using this data, answer the following questions:

   (a) Calculate the speedup, $\psi(n, p)$, for this program.
   (b) Calculate the efficiency, ϵ, for this program.
   (c) Calculate the sequential fraction of the program.
   (d) What is the value of speedup predicted by Amdahl's law when run using 100 processes?
   (e) What is the maximum speedup that can be obtained, as predicted by Amdahl's law, when run using infinite number of processes?

2. (20 points) Solve the following system of linear equations using recursive-doubling algorithm by clearly showing all the required steps. You may use the recursive expressions that we have developed or solve it directly using the philosophy of the method.

$$\begin{aligned} 3x_1 - x_2 &= 2 \\ -x_1 + 3x_2 - x_3 &= 1 \\ -x_2 + 3x_3 - x_4 &= 1 \\ -x_3 + 3x_4 &= 2. \end{aligned}$$

   Note: The answer to this trivial question can be obtained easily. You will get points only if you present your solution using recursive-doubling algorithm that we learnt in the class.

Programming Questions

1. (60 points) You are given a serial C code that performs Cholesky decomposition, which decomposes a symmetric positive definite matrix into a lower and an upper triangular matrix that are transposes of each other. Thus, given a matrix $A[N][N]$, it outputs a lower-triangular matrix $L[N][N]$ such that $A = LL^T$. The computation is specified as below.

$$L(i,j) = \begin{cases} i = j & : \sqrt{A(i,i) - \sum_{k=0}^{i-1} L(i,k)^2} \\ i > j & : \dfrac{A(i,j) - \sum_{k=0}^{j-1} L(i,k)L(j,k)}{L(j,j)} \end{cases} \tag{1}$$

The given serial code performs minor adjustments to reduce chances of overflows and underflows. Parallelize the given code using OpenACC ensuring correctness. You are free to modify the code. Optimize the OpenACC parallel code that you develop for data transfer, parallelism, and number of gangs. Plot the time taken by the serial and the parallel codes for N = 10, 100, 1000 in the same plot. For $N = 10$ show and ensure that the lower triangular matrix obtained is the same between serial and parallel solutions.